

# WCAG 2.1 AA Accessibility Audit

## 1. Accessibility Testing Methodology

### Automated Testing

The WAVE accessibility evaluation tool was used on each of the five pages in the evaluated workflow.

The tool identified:

- Accessibility errors
- Color contrast problems
- Accessibility alerts
- Alternative-text issues
- Form-label issues
- Heading-structure issues
- ARIA-related information
- Structural and semantic information

WAVE results were treated as evidence of potential accessibility issues rather than automatic proof of WCAG failure, because automated tools cannot determine every accessibility requirement on their own.

### Manual Testing

Manual keyboard testing was performed using the **Tab**, **Enter**, and **Space** keys.

The evaluation examined whether:

- Interactive elements could be reached with the keyboard.
- The current keyboard focus was visible.
- Focused elements could be activated.
- Important controls could be reached without using a mouse.

## 2. WAVE Automated Testing Results

The WAVE results showed significant accessibility issues across the Amazon pages evaluated.

| Page | Errors | Contrast Errors | Alerts | AIM Score |
|------|--------|-----------------|--------|-----------|
|------|--------|-----------------|--------|-----------|

|                    |     |     |     |        |
|--------------------|-----|-----|-----|--------|
| Shopping Cart      | 5   | 82  | 190 | 3.1/10 |
| Checkout           | 3   | 21  | 9   | 6.5/10 |
| Product Page       | 113 | 138 | 422 | 2.8/10 |
| Amazon/Search Page | 0   | 189 | 203 | 1.9/10 |
| Search Results     | 498 | 265 | 115 | 1.4/10 |

The results demonstrate that accessibility issues were not isolated to one portion of the website. In particular, contrast errors occurred on every evaluated page, while the search-results page produced the largest number of reported errors.

The WAVE results should be interpreted in context because some reported items may represent warnings, repeated elements, or implementation details rather than independently meaningful WCAG violations.

### 3. WCAG 1.4.3 — Contrast

**Level:** AA

**Severity:** High

#### Finding

Color contrast was the most consistently identified accessibility concern across the evaluated Amazon pages.

WAVE reported:

- **82 contrast errors** on the shopping cart.
- **21 contrast errors** on the checkout page.
- **138 contrast errors** on the product page.
- **189 contrast errors** on the Amazon/search page.
- **265 contrast errors** on the search-results page.

The presence of contrast errors across all five evaluated pages indicates a recurring accessibility concern rather than an isolated problem.

#### Why This Matters

Users with low vision or color-vision deficiencies may have difficulty distinguishing text or interface elements when there is insufficient contrast between foreground and background colors.

This is particularly important on an e-commerce website because users need to distinguish:

- Prices
- Product information
- Links
- Buttons
- Delivery information
- Filters
- Navigation elements
- Promotional information

If these elements are difficult to perceive, users may have difficulty completing a purchase.

## **Evidence**

The WAVE screenshots visually identify numerous low-contrast elements throughout the evaluated pages.

## **Severity**

### **3 — Major usability/accessibility problem**

The problem can interfere with users' ability to perceive important information and controls, although it does not prevent every user from completing the task.

## **Recommendation**

Amazon should review the identified text and interface elements against the WCAG contrast requirements and increase contrast where necessary. Contrast should be evaluated for both normal text and user-interface components rather than relying only on the overall visual appearance of the page.

## **4. WCAG 1.1.1 — Non-text Content**

**Level:** A

**Severity:** Moderate–High

## **Finding**

WAVE identified missing alternative text on several evaluated pages.

Examples from the WAVE results include:

- Missing alternative text on the cart page.
- Missing alternative text on the product page.
- Missing alternative text on search results.
- Missing alternative text associated with images and other non-text content.

The search-results page produced particularly large numbers of reported errors, including 5 missing-alternative-text errors visible in the WAVE results.

## **Why This Matters**

Images are a major component of Amazon's shopping experience. Users relying on screen readers may need alternative text to understand what products are being displayed.

For example, a product image should communicate meaningful information such as the product name or relevant visual description when that information is necessary to understand the page.

Without appropriate alternative text, a screen-reader user may encounter an image without sufficient information about the product it represents.

## **Severity**

### **3 — Major accessibility concern**

Missing alternative text can prevent users who cannot see images from obtaining important product information.

## **Recommendation**

Amazon should ensure that meaningful product and interface images have appropriate alternative text. Decorative images should instead be correctly identified as decorative so that screen readers do not unnecessarily announce them.

## **5. WCAG 3.3.2 — Labels or Instructions**

**Level:** A

**Severity:** Moderate

## **Finding**

WAVE identified form-label issues on the evaluated pages.

The product-page WAVE results, for example, reported:

- **2 missing form labels**
- **7 empty form labels**

Other pages also contained form and control-related accessibility warnings.

## **Why This Matters**

Users who rely on screen readers need meaningful labels to understand what a form control does.

A visually obvious control may not provide enough information to a screen-reader user if the control does not have an appropriate accessible name.

This can affect controls such as:

- Search fields
- Quantity controls
- Filters
- Delivery selections
- Account-related controls

## **Severity**

### **2 — Moderate usability problem**

The impact depends on which controls are affected. Missing labels on important purchasing controls could become a major barrier.

## **Recommendation**

All interactive form controls should have clear, programmatically associated labels or accessible names. Empty labels should be removed or corrected rather than leaving assistive technologies with incomplete information.

# **7. WCAG 2.1.1 — Keyboard**

**Level:** A

**Severity:** 0 — No major issue observed

## **Finding**

Manual testing demonstrated that the evaluated Amazon workflow could be navigated using the keyboard.

The **Tab** key successfully moved focus between interactive elements.

The focused element was visually identified with a clear outline.

The **Enter** key successfully activated the focused element.

## Observed Behavior

| Keyboard Action | Result                                            |
|-----------------|---------------------------------------------------|
| <b>Tab</b>      | Successfully moved through interactive elements   |
| Focus indicator | Clearly visible                                   |
| <b>Enter</b>    | Activated focused item                            |
| <b>Space</b>    | Moved page rather than activating the tested link |

The behavior of the Space key was not considered a WCAG failure by itself because the tested elements were standard links, for which **Enter** is the expected activation mechanism.

## Severity

**0 — No issue observed**

## Recommendation

No immediate corrective action is required based on the tested workflow. However, Amazon should continue ensuring that custom interactive controls follow expected keyboard interaction patterns.

# 8. WCAG 2.4.7 — Focus Visible

**Level:** AA

**Severity:** 0 — No issue observed

## Finding

Keyboard testing showed that the currently focused element had a visible focus indicator.

For example, when navigating through Amazon's navigation using the **Tab** key, the currently focused "**Buy Again**" navigation element displayed a visible rectangular outline.

This allowed the evaluator to determine which interface element currently had keyboard focus.

## Why This Matters

Visible focus is particularly important for users who navigate websites without a mouse.

Without a visible focus indicator, a keyboard user could lose track of their location on the page and have difficulty interacting with the correct control.

## Severity

**0 — No issue observed**

## Recommendation

Amazon should maintain the current visible focus treatment and ensure that the focus indicator remains sufficiently distinguishable across all interactive components.

# 9. Semantic Structure and ARIA

**Severity:** Moderate–High

## Finding

WAVE identified a substantial amount of ARIA and structural information throughout the Amazon interface.

The WAVE results showed elements containing attributes such as:

- `aria-label`
- `aria-labelledby`
- `aria-describedby`
- `aria-controls`
- `tabindex`
- ARIA roles

The tool also identified structural concerns including skipped heading levels, empty headings, empty buttons, and other semantic issues.

For example, the evaluated pages contained WAVE alerts for:

- Skipped heading levels
- Empty headings
- Empty buttons

- Redundant links
- Empty form labels
- ARIA-related implementation

## Why This Matters

ARIA can improve accessibility when implemented correctly, but incorrect or unnecessary ARIA can make an interface more difficult for assistive technologies to interpret.

Amazon's interface is highly dynamic and contains a large number of interactive components, making consistent semantic structure particularly important.

## Severity

### 2 — Moderate

Not every WAVE structural or ARIA finding represents a user-facing WCAG violation, but the volume of reported issues indicates that the underlying accessibility implementation should be reviewed.

## Recommendation

Amazon should:

- Maintain a logical heading hierarchy.
- Remove empty headings and buttons.
- Ensure ARIA labels accurately describe controls.
- Avoid unnecessary ARIA when native HTML semantics are sufficient.
- Verify custom controls with actual screen-reader testing.

# 10. Key Accessibility Findings

The audit identified four primary areas requiring attention.

## Finding 1 — Insufficient Color Contrast

**WCAG:** 1.4.3

**Severity:** 3 — Major

Contrast errors appeared across all five tested pages, with the search-results page producing **265 reported contrast errors**.

## Finding 2 — Missing Alternative Text

**WCAG:** 1.1.1

**Severity:** 3 — Major

WAVE identified missing alternative text associated with images and other non-text content.

This is particularly significant for Amazon because product images are central to the shopping experience.

## **Finding 3 — Form and Control Labeling**

**WCAG:** 3.3.2

**Severity:** 2 — Moderate

WAVE identified missing and empty form labels. These issues may make certain controls difficult for screen-reader users to understand.

## **Finding 4 — Keyboard Focus**

**WCAG:** 2.1.1 / 2.4.7

**Severity:** 0 — No issue observed

The tested workflow demonstrated functional keyboard navigation and visible focus indicators. This represents an area where the evaluated interface performed well.

# **11. Accessibility Impact on the User Experience**

The accessibility findings have implications beyond users who identify as having a disability.

Poor contrast can affect anyone using a dim or bright display, experiencing screen glare, or viewing the website on a low-quality monitor.

Missing alternative text creates a significant barrier for blind and low-vision users but can also affect users when images fail to load.

Poor labeling can increase cognitive effort for all users, particularly when Amazon presents a large number of controls and purchasing options.

Therefore, improving accessibility would not only benefit users with disabilities but could also make the shopping experience clearer and easier for the broader user population.

# **12. Overall Assessment**

Based on the automated WAVE testing and manual keyboard evaluation, the Amazon shopping workflow demonstrates strong basic keyboard interaction but substantial accessibility concerns in visual contrast, alternative text, and semantic implementation.

The manual keyboard test did not identify a major focus-visibility problem. However, the WAVE results demonstrate that accessibility quality varies considerably between different pages.

The search-results page produced the largest number of reported errors, while the checkout page produced substantially fewer errors. This suggests that accessibility should be evaluated at the individual page and component level rather than assuming that accessibility performance is consistent throughout the entire Amazon platform.

It is also important to note that WAVE is an automated screening tool. Its results should be combined with manual testing and, ideally, testing with assistive technologies such as screen readers before making a complete WCAG conformance claim.

## **13. Audit Conclusion**

The accessibility audit identified several areas where Amazon could reduce barriers for users with disabilities.

The most significant concerns observed during this evaluation were color contrast and missing alternative text, followed by issues involving form labeling and semantic structure.

At the same time, the manual keyboard evaluation demonstrated that Amazon provides a visible focus indicator and allows the tested interactive elements to be reached and activated with the keyboard.

Overall, the findings indicate that Amazon's interface provides a functional foundation for keyboard users but has opportunities to improve its accessibility implementation, particularly for users with visual and screen-reader-related accessibility needs.